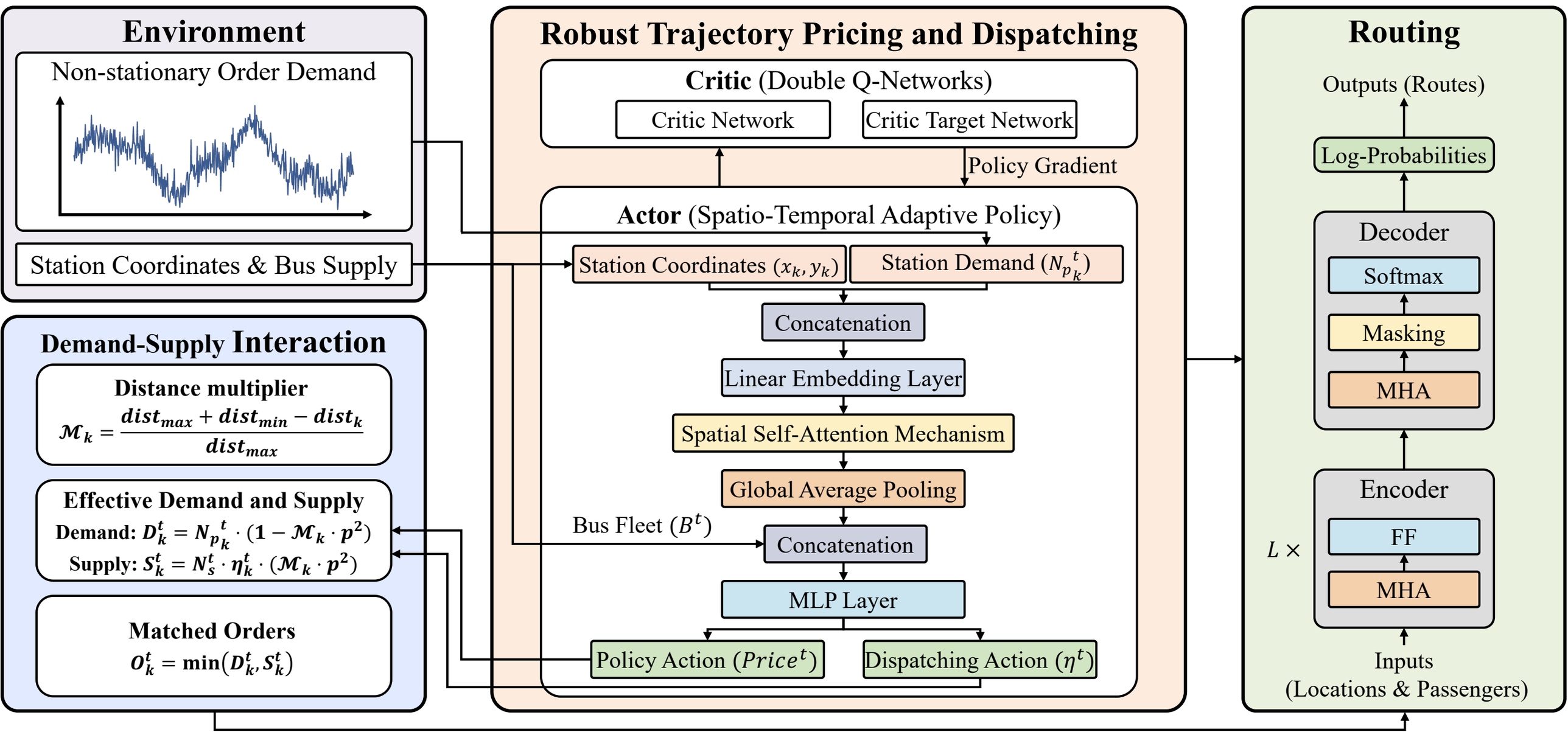

Environment
Non-stationary Order Demand
Station Coordinates & Bus Supply
Demand-Supply Interaction
Distance multiplier
$\mathcal{M}_k = \frac{dist_{max} + dist_{min} - dist_k}{dist_{max}}$
Effective Demand and Supply
Demand: $D_k^t = N_{p_k}^t \cdot (1 - \mathcal{M}_k \cdot p^2)$
Supply: $S_k^t = N_s^t \cdot \eta_k^t \cdot (\mathcal{M}_k \cdot p^2)$
Matched Orders
$O_k^t = \min(D_k^t, S_k^t)$
Robust Trajectory Pricing and Dispatching
Critic (Double Q-Networks)
Critic Network
Critic Target Network
Policy Gradient
Actor (Spatio-Temporal Adaptive Policy)
Station Coordinates (x_k, y_k)
Station Demand $(N_{p_k}^t)$
Concatenation
Linear Embedding Layer
Spatial Self-Attention Mechanism
Global Average Pooling
Bus Fleet (B^t)
Concatenation
MLP Layer
Policy Action $(Price^t)$
Dispatching Action (η^t)
Routing
Outputs (Routes)
Log-Probabilities
Decoder
Softmax
Masking
MHA
$L \times$
Encoder
FF
MHA
Inputs
(Locations & Passengers)